GROUP 9

ALGORITHM, FLOWCHARTS & PSEUDOCODE

ALGORITHM RUSH!!!

Arrange the given steps in the correct order to complete the task successfully.

How To Play:

1. Read all the given steps carefully.
2. Identify which step should come first.
3. Arrange the steps in the correct sequence.
4. Make sure each step logically follows the previous one.
5. Choose the number corresponding to your answer.

MAKING COFFEE CHALLENGE

Can you arrange the following steps

1. Stir the coffee.

2. Drink the coffee.

3. Get a cup.

4. Pour hot water into the cup

5. Add coffee and sugar.

1.Boil water

7. Add milk

__ __ __ __ __ __ __

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6 3 5 4 7 1 2



MAKING A TOAST

Can you arrange the following steps

1. Put the toast on a plate.

1.Put bread into the toaster.

3. Apply butter.

4. Get a slice of bread.

5. Press the lever.

6. Set the toasting level.

7. Plug in the toaster.

8. Wait for the toast to pop up.

— — — — — — — —



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4 2 7 6 5 8 1 3



SUBMITTING AN ASSIGNMENT

Can you arrange the following steps

1. Upload the file.

2. Check if the file is correc

3. Open the online classroom.

4. Select the assignment.

5. Log in to your account.

6. Select the file.

7. Click Submit.

8. Check if the submission was successful.

— — — — — — — —



SUBMITTING AN ASSIGNMENT

Can you arrange the following steps

1. Upload the file.

2. Check if the file is correct

3. Open the online classroom.

4. Select the assignment.

5. Log in to your account.

6. Select the file.

7. Click Submit.

8. Check if the submission was successful.

5 3 4 6 2 1 7 8



ORDERING FOOD ONLINE

Can you arrange the following steps

1. Add the foods to your cart.

2. Open the delivery app.

3. Confirm the order.

4. Check the total price.

5 Choose your food.

6 Enter your address.

7 Select your payment method.

8 Wait for the delivery.

__ __ __ __ __ __ __ __



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8 Wait for the delivery.

2 5 1 4 6 7 3 8

FIND THE MISSING FILE

Can you arrange the following steps

1.Close the File Explorer

2. Verify that the file is the correct one.

3. Check the Documents folder.

4. Move the file to the correct folder.

5. Search for the file name.

6. Open File Explorer.

7. Open the file if found.

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CORE PRINCIPLE

PLANNING BEFORE PROGRAMMING

Jumping straight into code often leads to fragile logic, high bug rates, and costly rewrites. Thinking twice lets you code once.

The Value of Pre-Planning

The 1:10 Engineering Rule :

Every hour invested in proper pre-coding logic planning prevents up to 10 hours of chaotic debugging, refactoring, and emergency hotfixes during production.

Flawed logic discovered at the blueprint stage costs nearly nothing to fix compared to flaws found in written code.

The Three Pre-Coding Pillars



1.ALGORITHMS



1.FLOWCHARTS



1.PSEUDO CODES

The Three Pre-Coding Pillars



ALGORITHMS

An algorithm is a step-by-step procedure used to solve a specific problem or complete a task. It provides a clear and logical sequence of instructions, starting from the given input and leading to the desired output. Algorithms help programmers plan how a problem should be solved before writing the actual code.

Algorithms:

Purpose

- Breaks a problem into manageable steps
- Provides a logical plan before coding
- Helps identify the required inputs and outputs

Characteristics

- Clear and unambiguous
- Has a definite beginning and end
- Steps are arranged in a logical order
- Should produce a desired result

Simple Sample : ALGORITHMS

Problem: Find the average of two numbers

STEP	PROCESS	RESULT
1. Input	Enter two numbers	10, 20
1. Add	10 + 20	30
1. Divide	30 ÷ 2	15
1.Output	Display the average	15

Simple Sample : ALGORITHMS

Step 1: Start

Step 2: Create a variable to receive the user's email address

Step 3: Clear the variable in case it's not empty

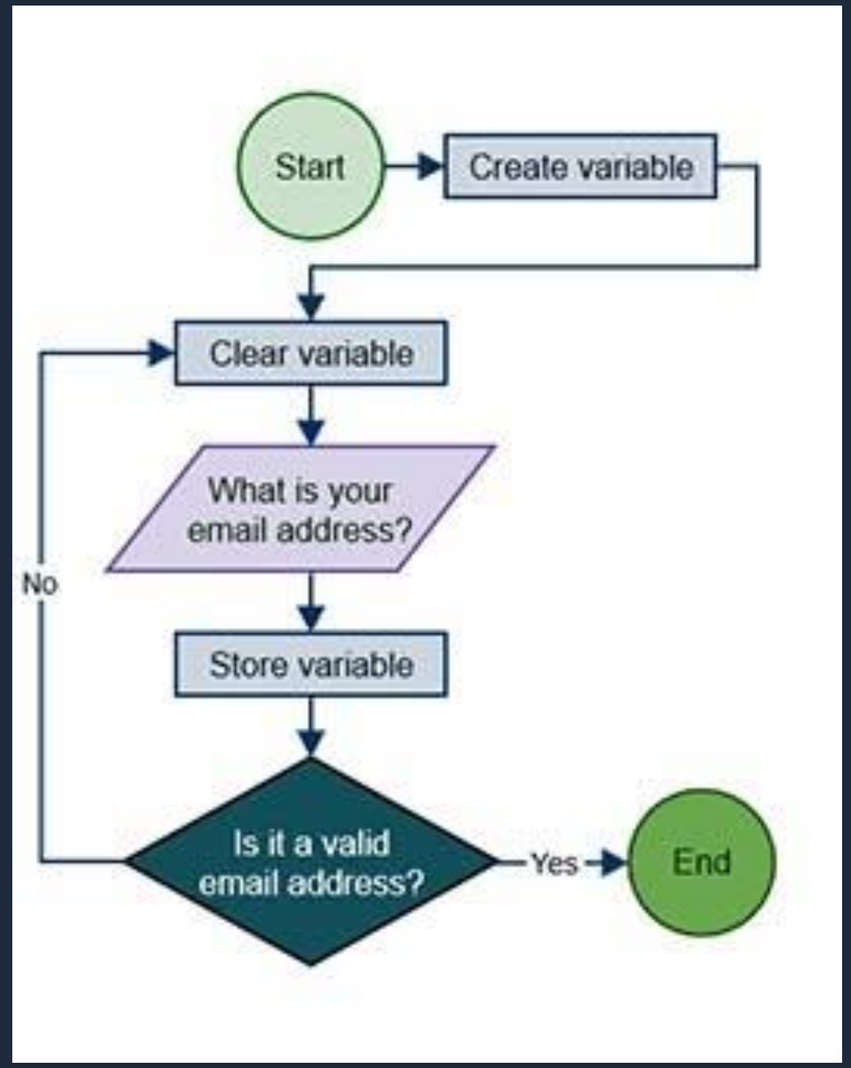
Step 4: Ask the user for an email address

Step 5: Store the response in the variable

Step 6: Check the stored response to see if it is a valid email address

Step 7: Not valid? Go back to Step 3.

Step 8: End



The Three Pre-Coding Pillars



FLOWCHARTS

A flowchart is a type of diagram that represents an algorithm or process using symbols, shapes, and arrows. It visually shows the sequence of steps and decisions involved in solving a problem, making the logic easier to understand and follow.

Flowcharts :

Purpose

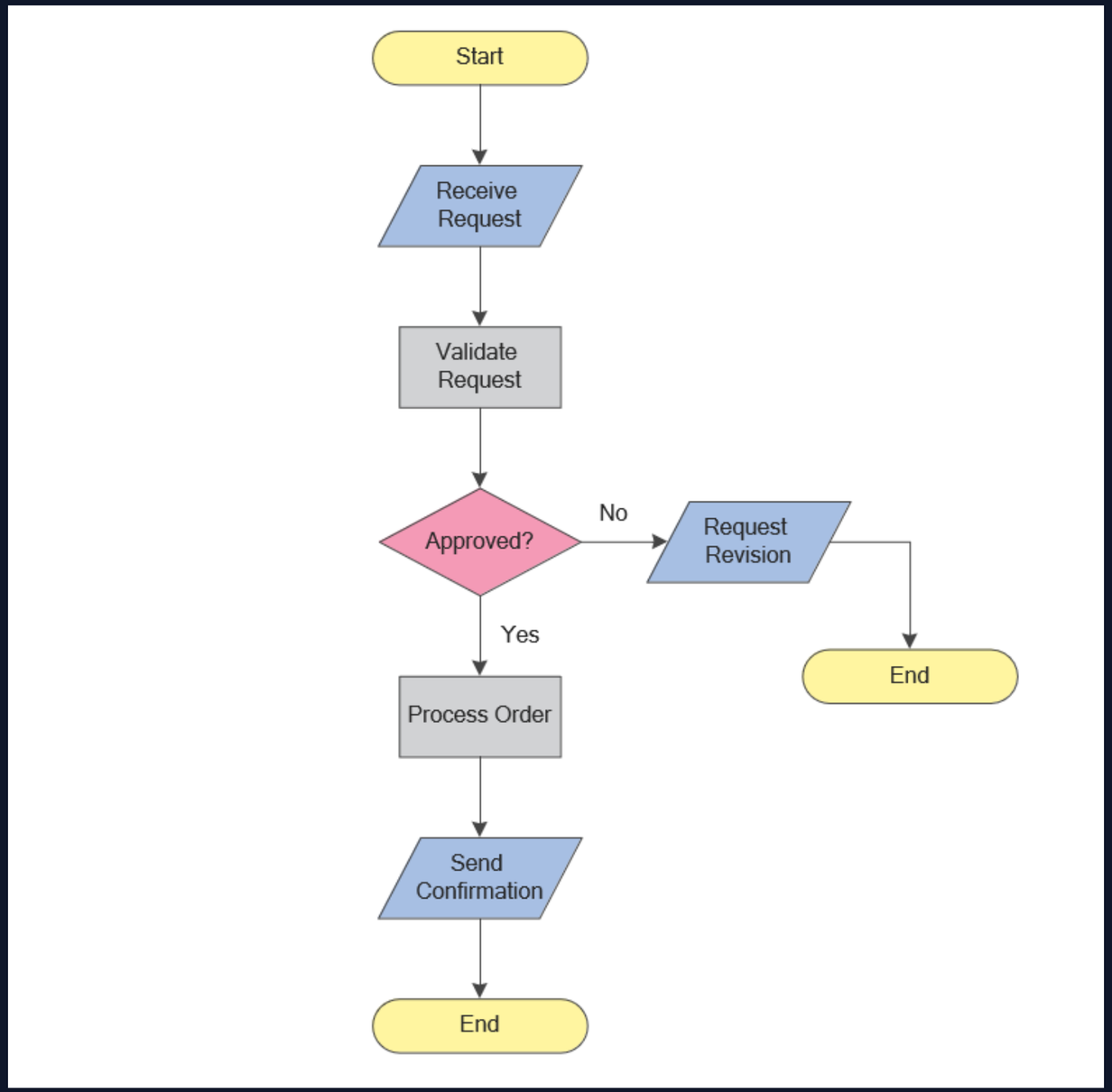
- Makes program logic easier to visualize
- Shows the sequence of operations
- Helps identify decisions and possible paths
- Makes errors in logic easier to spot

Standard Flowchart Symbols

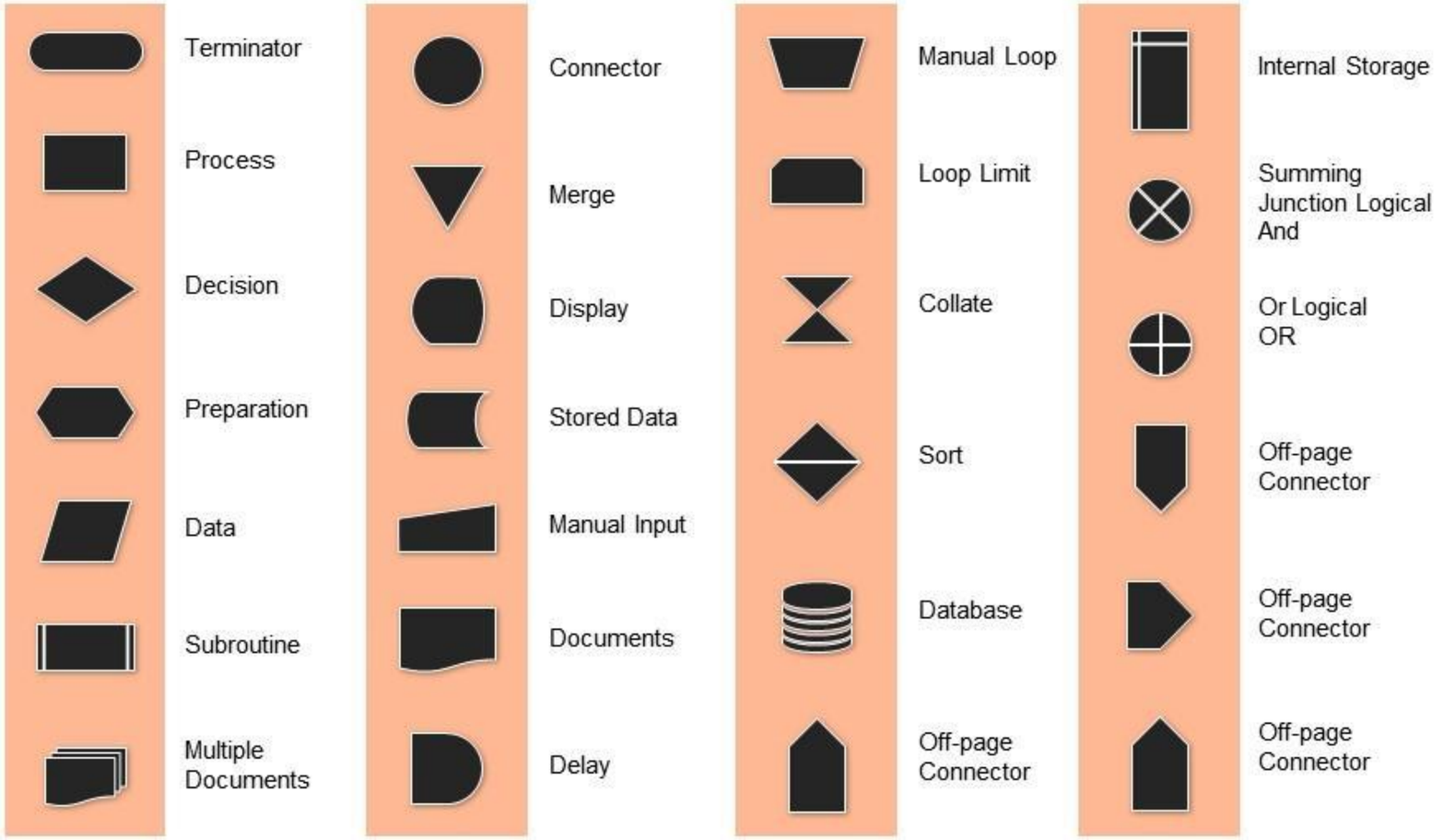
- **Terminator (Oval):** Defines the absolute START and END points of an execution path.
- **Process (Rectangle):** Represents action steps, calculations, or variable initializations.
- **Decision (Diamond):** Evaluates TRUE/FALSE conditions that branch program execution.
- **Flowline (Arrow):** A line is a connector that shows relationships between the representative shapes.
- **Input / Output (Parallelogram):** A parallelogram represents input and output

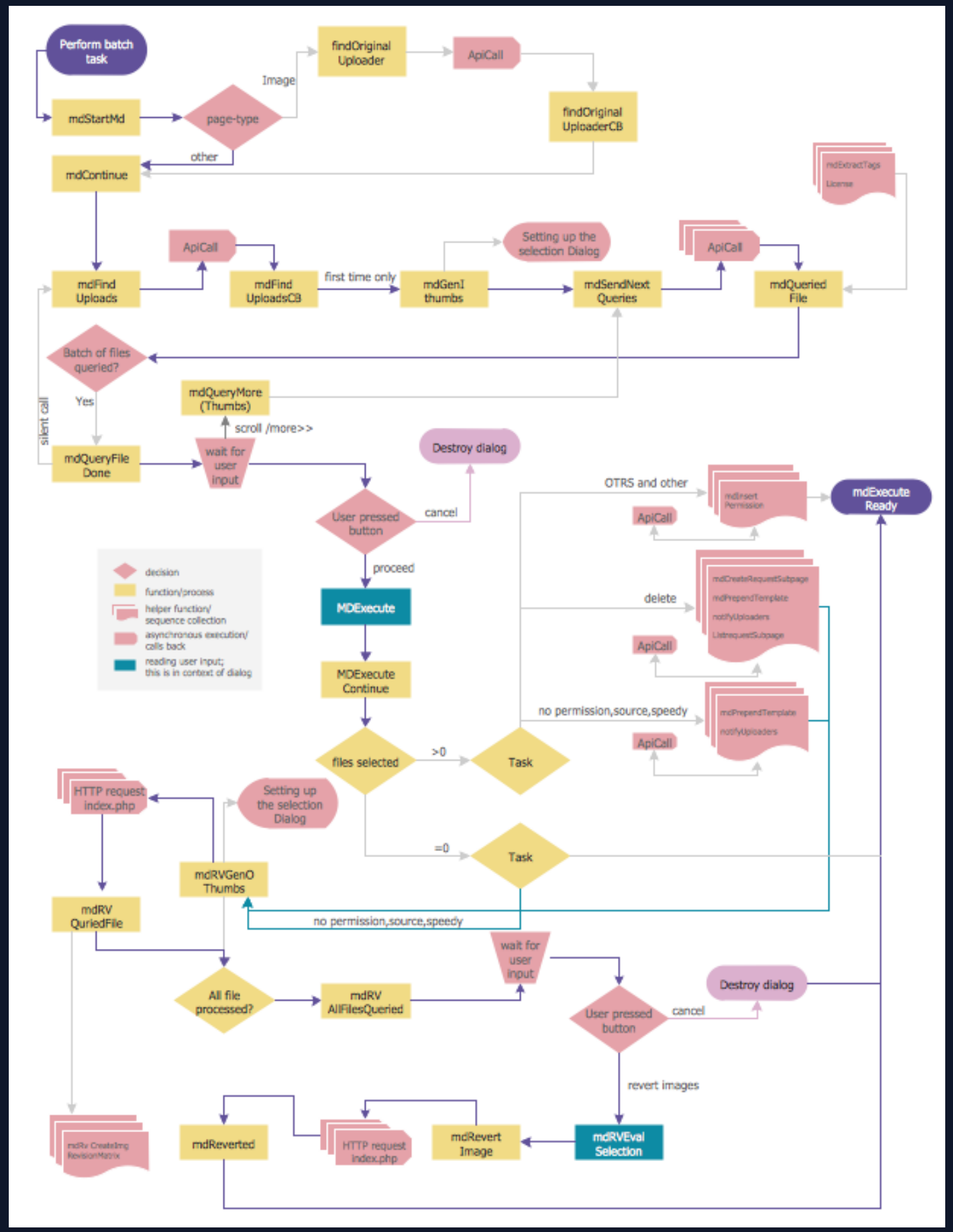
Standard Flowchart Symbols

- **Preparation / Initialization (Hexagon):** Represents a preparation or initialization step, such as setting initial values or preparing variables.
- **On-Page Connector (Circle):** Connects different parts of a flowchart on the same page. Matching labels show where the flow continues.
- **Off-Page Connector (Pentagon):** Indicates that the flowchart continues on another page.


Start
Receive Request
Validate Request
Approved?
No
Request Revision
End
Yes
Process Order
Send Confirmation
End

Complex Process Flow Chart Symbols




Perform batch task
findOriginal Uploader
ApiCall
Image
mdStartMd
page-type
findOriginal UploaderCB
other
mdContinue
mdExtractTags
License
ApiCall
Setting up the selection Dialog
ApiCall
mdFind Uploads
mdFind UploadsCB
first time only
mdGenI thumbs
mdSendNext Queries
mdQueried File
Batch of files queried?
mdQueryMore (Thumbs)
Yes
silent call
scroll /more>>
mdQueryFile Done
wait for user input
Destroy dialog
OTRS and other
mdExecute Ready
User pressed button
cancel
ApiCall
mdInsert Permission
proceed
decision
function/process
helper function/ sequence collection
asynchronous execution/ calls back
reading user input; this is in context of dialog
MDExecute
delete
mdCreateRequestSubpage
mdPrependTemplate
notifyUploaders
ListrequestSubpage
ApiCall
MDExecute Continue
no permission,source,speedy
mdPrependTemplate
notifyUploaders
ApiCall
files selected
>0
Task
HTTP request index.php
Setting up the selection Dialog
=0
Task
mdRVGenO Thumbs
mdRV QuriedFile
no permission,source,speedy
wait for user input
All file processed?
mdRV AllFilesQueried
Destroy dialog
User pressed button
cancel
revert images
mdRv CreateImg RevisionMatrix
mdReverted
HTTP request index.php
mdRevert Image
mdRVEval Selection

The Three Pre-Coding Pillars



PSEUDO CODE

A simple way to plan a computer program. It mixes plain human words with structural programming keywords. It describes what code should do without using strict computer syntax.

Pseudocode: Human-Readable Code

Structured Plain Text

Pseudocode bridges natural language and formal programming languages using clean syntax constructs like `IF-ELSE`, `WHILE` and `FOR`.

It allows developers to review algorithms together before typing syntax into an IDE.

Pseudocode: Human-Readable Code

```
// Example Pseudocode
BEGI  ProcessUserLogin
N INPU  username, password
  I  ValidateCredentials(username, password) THE
  F DISPLA  "Welcome Back!"                  N
  ELS
  E DISPLA  "Access Denied"
  END
ENF
D
```

Pseudocode: Human-Readable Code

```
BEGIN

  SET count = 5

  WHILE count > 0

    OUTPUT count

    count = count - 1

  END WHILE

  OUTPUT "Blast off!"

END
```

Pseudocode: Human-Readable Code

```
START

IF it is raining THEN
    Take umbrella
ELSE
    Enjoy the sun
END IF

END
```

Technique Comparison Matrix

Tool	Primary Purpose	Best Audience	Key Advantage
	Step-by-step solution	Programmers/Students	Language-independent
	Visualizing processes and decisions	Understanding logic	Easy visual representation
	Planning programming structure	Preparing for coding	Simple and readable
Algorithm			
Flowchart			
Pseudocode			

Pre-Coding Workflow



Deconstruct

Analyze inputs, desired outputs, and edge cases.

Algorithm

Draft high-level step-by-step logic rules.

Flowchart

Build flowchart to verify decision paths.

Pseudocode

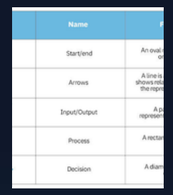
Draft structural logic & review before coding.

Image and Reference Sources



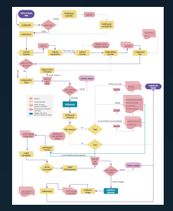
https://www.conceptdraw.com/How-To-Guide/flowchart-symbols

Source: www.conceptdraw.com



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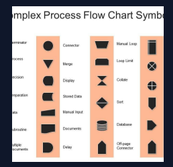
Source: www.processon.io



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Image and Reference Sources



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https://www.edukin.com/lessons/06-while-loops

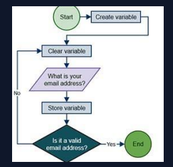
Source: www.edukin.com



https://www.codingireland.ie/ViewLesson/flowcharts-and-pseudocode-9714/5441

Source: www.codingireland.ie

Image and Reference Sources



https://study.com/academy/lesson/what-is-an-algorithm-in-programming-definition-examples-analysis.html

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